SMARTBRIDGE
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a Veranda Enterprise

Smart Internz

Model Development Phase Template

Date	15 Nov 2024
Team ID	739938
Project Title	AI Enabled Candidate Resume Screening
Maximum Marks	5 Marks

Model Selection Report

The Model Selection Report for AI-enabled candidate resume screening using spaCy models focuses on the integration of natural language processing (NLP) techniques to enhance the efficiency and accuracy of the recruitment process. AI resume screening automates the process of reviewing and ranking job applicants' resumes. By leveraging machine learning and NLP, these systems can quickly analyze large volumes of resumes, identify key skills, and match candidates to job descriptions more effectively than traditional methods.

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Model Selection Report:

Model	Description
Model 1	spaCy is a powerful, open-source library designed for advanced natural language processing (NLP) in Python. It was developed by Matthew Honnibal and Ines Montani, and first released in 2015. Unlike other NLP libraries such as NLTK, which are often used for educational purposes, spaCy focuses on providing tools suitable for production environments, making it ideal for building applications that require efficient text processing and understanding. Key Components:- **High Performance**: spaCy is optimized for speed and efficiency, allowing it to handle large volume of text quickly.

	Pre-trained Models: The library includes pre-trained statistical models for a variety of languages, supporting tasks such as part-of-speech tagging, dependency parsing, named entity recognition (NER), and text classification. Currently, it offers models for 23 languages, each equipped with the necessary resources for these NLP tasks. **Transformer Integration:** Starting with version 3.0, spaCy introduced support for transformer-based models such as BERT and GPT. This enables users to leverage state-of-the-art NLP techniques, significantly improving the accuracy and performance of their applications. **Custom Model Training:** spaCy provides a flexible training framework that allows users to train custom models using their own datasets. It seamlessly integrates with popular deep learning libraries like TensorFlow and PyTorch, offering robust support for advanced NLP workflows. **Built-in Visualizations:** The library includes built-in tools like displaCy for visualizing dependency parse trees and named entities, making it easier to interpret and debug model outputs.